CARECONNECT

Donation & Beneficiary Management System

Presented By: Krishna Chaurashia
Niraj Naik
Anurag Teurwade
Virendra Khade

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PROBLEM STATEMENT

- Traditional donation systems lack transparency and verification
- Manual handling of beneficiary requests causes delays
- Difficult to manage donors, beneficiaries, and approvals
- Security risks in role-based access
- Poor scalability in monolithic systems

Solution

CareConnect provides a **secure, scalable microservices-based platform** to connect **donors, beneficiaries, and admins** efficiently.

USERS

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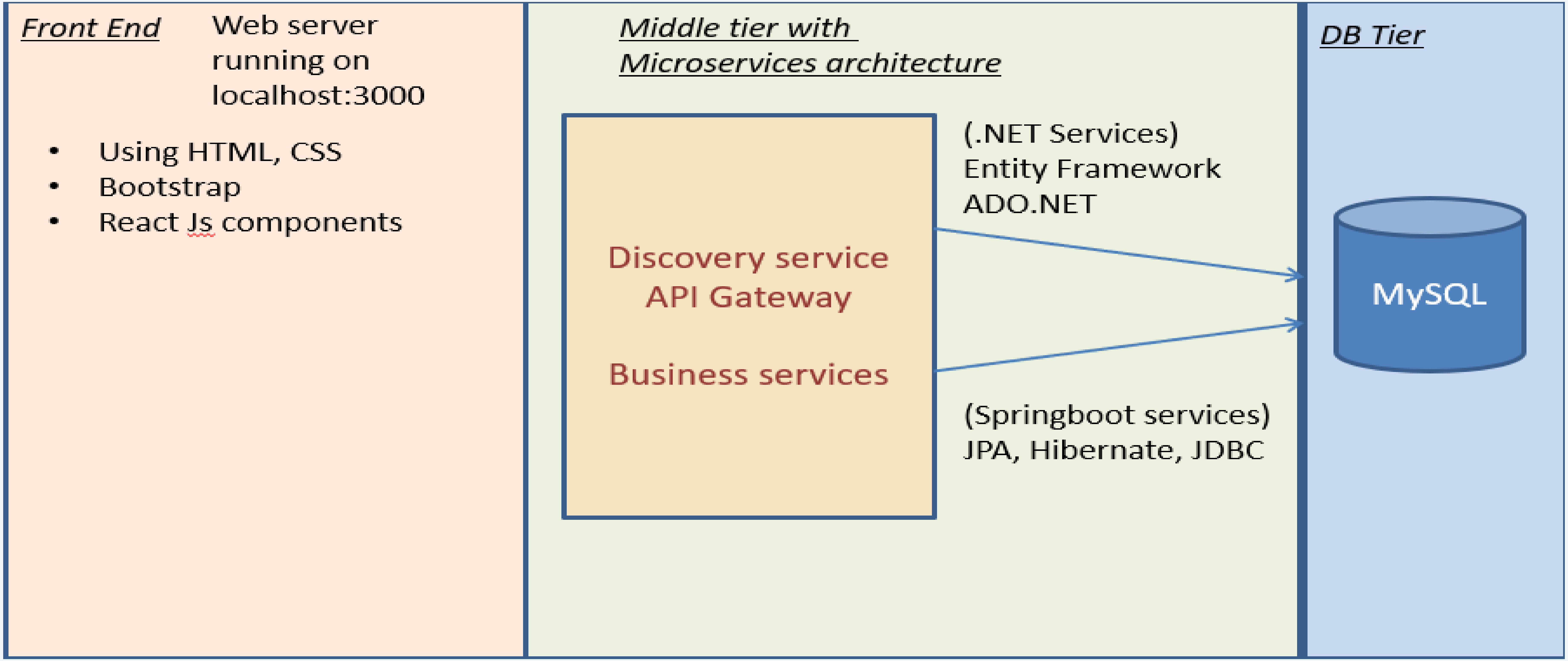
The CareConnect platform supports **role-based access control**:
Admin
Beneficiary
Donor:
Individual Donor
Organization Donor

- Donors browse verified requests and donate funds/resources
- Beneficiaries submit and track aid requests
- Admin verifies users, approves requests, and monitors transactions
- System maintains audit logs for accountability

PROJECT ARCHITECTURE



PROJECT ARCHITECTURE

- **Architecture Type:**
- Role-based, modular web application
  Layered architecture
- **Main Components:**
- **Frontend:** User interfaces for Admin, Donor, and Beneficiary
  Backend: REST APIs handling business logic
- **Database:** Relational database for secure data storage (MySQL)
- **Key Architectural Features:**
- Separation of concerns (UI, business logic, data)
  Secure authentication and authorization
  Scalable and maintainable design

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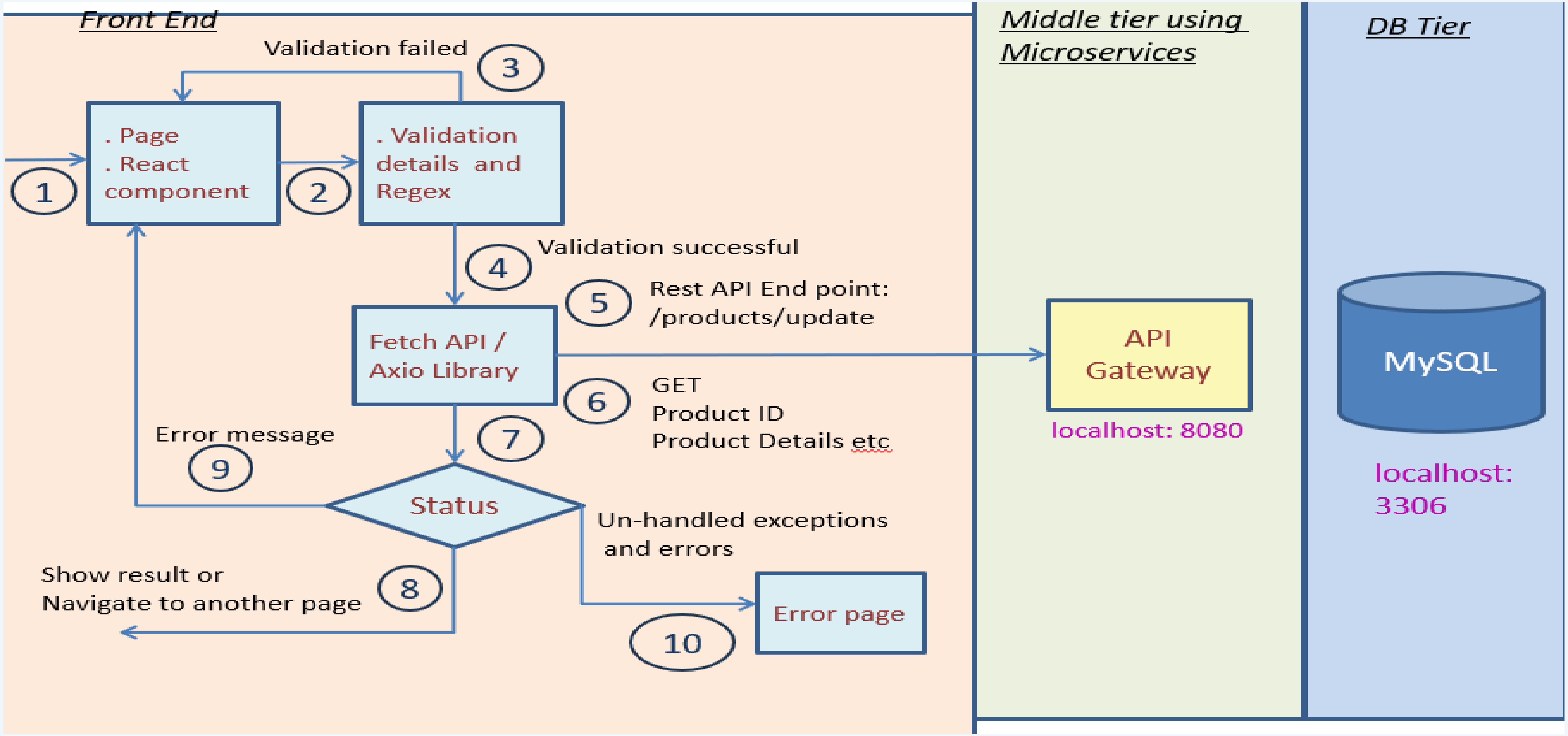
FRONTEND WORKFLOW



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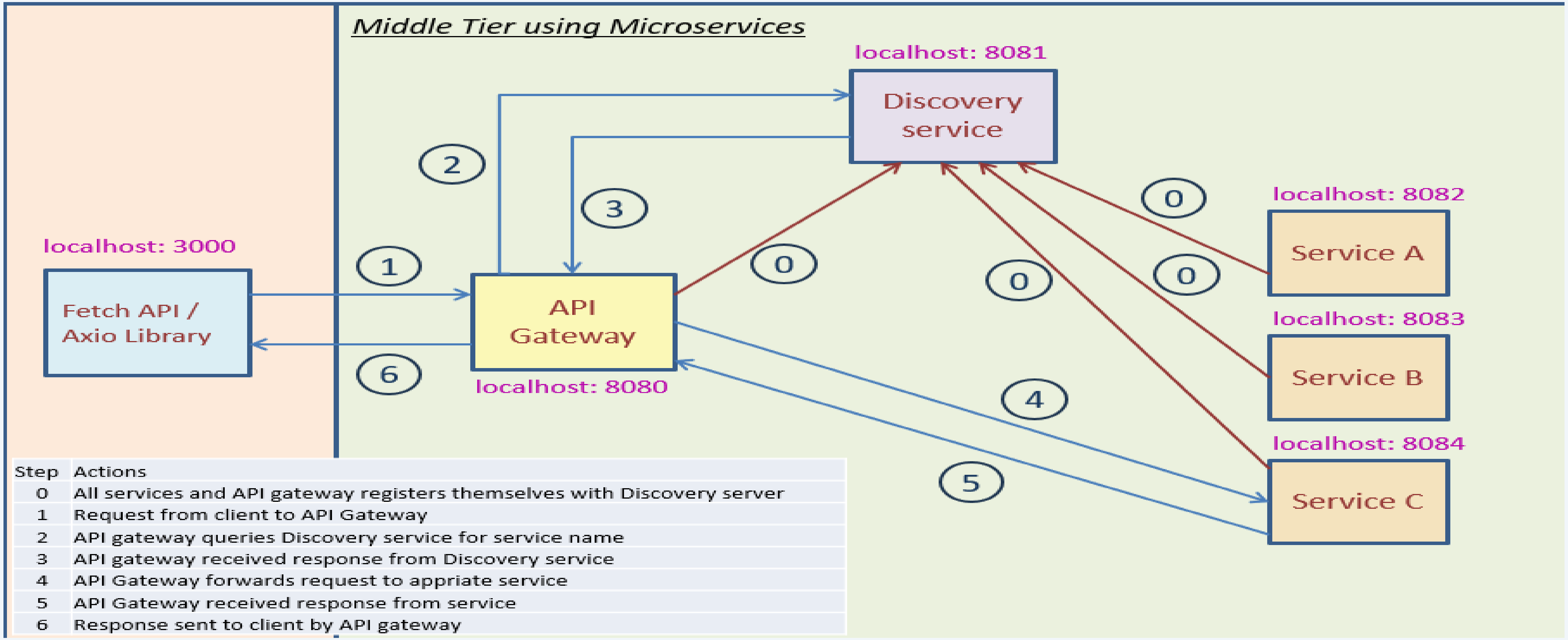
FRONTEND WORKFLOW

- User Interact with Ui
- Client-side Validations
- API calls using Axios
- JWT call using Axios
- Backend Processes Request
- Response displayed in UI

MICROSERVICES ARCHITECTURE AND WORKFLOW



Step	Actions
0	All services and API gateway registers themselves with Discovery server
1	Request from client to API Gateway
2	API gateway queries Discovery service for service name
3	API gateway received response from Discovery service
4	API Gateway forwards request to appriate service
5	API Gateway received response from service
6	Response sent to client by API gateway

MICROSERVICES ARCHITECTURE AND WORKFLOW

Microservices Architecture – CareConnect
CareConnect is designed using a **Microservices Architecture** to ensure scalability, flexibility, and maintainability.
Each service is developed, deployed, and scaled **independently**.
The system follows a **three-tier architecture**:
Frontend (React.js)
Middle Tier (Microservices)
Database Layer (MySQL)

Services Used in CareConnect
Auth Service (Spring Boot)
Handles authentication, authorization, and JWT token generation.
Beneficiary Service (Spring Boot)
Manages beneficiary profiles, help requests, and verification status.
Donor Service (.net)
Handles donor registration and donation management
Admin Service (Spring Boot)
Maintains verification logs and approval workflows.
API Gateway
Acts as a single entry point for all client requests.

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PROJECT METHODOLOGY

- Requirements were analyzed and divided into independent services such as Donor ,Beneficiary, and Admin.
- The system was designed using a three-tier architecture: Frontend, Microservices, and Database.
- Microservices were developed using Spring Boot and ASP.NET Core with REST Template for communication.
- Each service uses shared database to ensure loose coupling and scalability.
- Frontend integration was done using React.js and Axios/Fetch API.
- Iterative testing and final integration ensured smooth end-to-end functionality.

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SERVICES

Admin Functionality
- Admin login with authentication
- Verify beneficiaries and donors
- Approve or reject help requests
- View verification logs
- Monitor overall system activity

Beneficiary Functionality
- Register and login securely
- Create and manage beneficiary profile
- Create help requests (food, medical, education, etc.)
- View request status (Open, Approved, Rejected, Fulfilled)
- View verification history

Donor Functionality
- Register and login securely
- View approved beneficiary requests
- Donate to beneficiaries
- Track donation history

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Workflow

--Beneficiary Workflow
Beneficiary registers and logs into the system
Creates and updates beneficiary profile
Raises help requests (Food, Medical, Education, etc.)
Requests are submitted for admin approval

– Admin Workflow
Admin logs into the system
Verifies beneficiary details and documents
Approves or rejects beneficiary requests
Approved requests are made visible to donors

– Donor Workflow
Donor registers and logs into the system
Views approved beneficiary requests
Selects a request and donates resources
Donation status is updated in the system

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My Experience / Lessons Learnt

Technical Learnings:
Real-world implementation of **role-based systems**
Database design with **foreign keys and constraints**
Secure API development and data handling
Professional Learnings:
Importance of requirement clarity
Team coordination and task ownership
Writing scalable and maintainable code
Overall Experience:
This project strengthened my **full-stack development skills**
Gave hands-on exposure to **real-world problem solving**
Improved confidence in designing end-to-end applications

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FUTURE EXTENSIONS

Mobile Application Development:-

Develop Android & iOS apps to provide on-the-go access for donor and beneficiary, Push notifications for Requests and reminders.

AI & Predictive Health Analytics:-

Intelligent matching of donor interests with beneficiary needs.

Automated Notifications & Reminders:-

Automated SMS/email/app notifications for requests.

Cloud Deployment & Scalability:-

Deploy on cloud for better performance, security, and scalability.

Easy integration with additional modules in future.

THANK YOU

Team CareConnect